Course Name: Computer Networks
Course Code: CSA0701
Student Name: Sai Aanandha Lakshmi R V
Registration Number: 1911260237

ASSIGNMENT-1

TASK 1 – SIZE THE TRAFFIC BEFORE CHOOSING ANYTHING

1. Introduction

Network traffic analysis is an important step in designing a reliable network for a smart manufacturing plant. Different devices generate different amounts of data and have different communication requirements. Robot controllers require extremely low latency, whereas CCTV cameras generate large volumes of continuous data. Therefore, the network must be designed to handle both high data volume and time-sensitive communication. The given manufacturing plant contains robot controllers, vibration sensors, CCTV cameras and office users. The traffic generated by these devices must be calculated before selecting the appropriate network infrastructure.

2. Calculation of Total Network Traffic

The traffic generated by each device category is calculated separately.

A. Robot Controllers

Number of controllers = 40

Data per command = 200 bytes

Command interval = 2 milliseconds = 0.002 seconds

Formula: Traffic = Number of devices × Data size × 8 / Time interval

= 40 × 200 × 8 / 0.002 = 32,000,000 bits/second

Robot traffic = 32 Mbps. Robot controllers generate frequent commands and require a very low delay to ensure accurate machine operation.

B. Vibration Sensors

Number of sensors = 600

Data per second = 50 bytes

Traffic = 600 × 50 × 8 = 240,000 bits/second

Sensor traffic = 0.24 Mbps. These sensors continuously monitor machine vibration and help identify abnormal operating conditions.

C. CCTV Cameras

Number of cameras = 24

Traffic per camera = 4 Mbps

Total traffic = 24 × 4 =96,000,000 bits/second

CCTV traffic = 96 Mbps. CCTV cameras generate the highest volume of continuous network traffic because they transmit video streams.

D. Office Users

Number of users = 150

Active users = 10% of 150 = 15 users

Traffic per active user = 2 Mbps

Total traffic = 15 × 2=30,000,000 bits/second

Office traffic = 30 Mbps. Only 10% of office users are assumed to be active simultaneously during peak usage.

Total Traffic of One Production Line

Source	Traffic
Robot controllers	32 Mbps
Vibration sensors	0.24 Mbps
CCTV cameras	96 Mbps
Office users	30 Mbps
Total	**158.24 Mbps**

Therefore, the total traffic generated by one production line is **158.24 Mbps**.

Traffic of the Entire Plant**-** The plant contains eight production lines.

Total plant traffic = 158.24 × 8 = 1265.92 Mbps = 1.266 Gbps approximately. Thus, the network must support approximately 1.266 Gbps of traffic when all eight production lines operate simultaneously.

3. Highest Data Source and Strictest Delay Requirement

CCTV cameras generate the highest amount of traffic at 96 Mbps per line. However, robot controllers have the strictest delay requirement because their commands must reach the machines within 10 milliseconds. Therefore, the highest-volume traffic source and the most time-sensitive traffic source are different. This distinction is important because a network designed only for high bandwidth may not provide sufficiently low latency for industrial control applications.

4. Evaluation of the 1 Gbps Uplink

A 1 Gbps uplink provides a maximum theoretical capacity of 1000 Mbps.

For one production line:

Traffic = 158.24 Mbps

Utilisation = (158.24 / 1000) × 100 = **15.824%**

Therefore, a 1 Gbps uplink is sufficient for one production line.

For eight production lines: Traffic = 1265.92 Mbps

Required capacity exceeds available capacity by: 1265.92 − 1000 = **265.92 Mbps**

Hence, a single 1 Gbps uplink is insufficient for the entire plant. A higher-capacity backbone, such as 2.5 Gbps or 10 Gbps, should be considered to accommodate traffic growth and additional network overhead.

Effect of CCTV Traffic on Robot Commands**-** If CCTV traffic and robot commands share the same queue, large video packets may occupy the transmission queue before smaller robot commands. This creates queueing delay and may cause robot commands to miss their 10 ms deadline. To avoid this problem, Quality of Service (QoS) should be implemented. Robot control traffic must receive the highest priority, while CCTV and office traffic can use lower-priority queues.

5. Ranking of Traffic According to the Effect of 100 ms Delay

Rank	Traffic	Effect of 100 ms delay
1	Robot controllers	Can violate the 10 ms deadline and affect machine operation.
2	Vibration sensors	May delay abnormal-condition detection.
3	CCTV cameras	Can cause noticeable video latency.
4	Office users	Usually experience only minor application delays.

Robot controllers are ranked first because their delay tolerance is much lower than 100 ms. Office applications generally tolerate comparatively higher delays.

6. Alternative Concept Analysis – Local CCTV Recording

An alternative approach is to record CCTV footage locally and transmit only motion-triggered clips. The cameras are assumed to transmit data for approximately 5% of the time.

Original CCTV traffic = 96 Mbps

New CCTV traffic = 96 × 5/100 = 4.8 Mbps

Recalculated Traffic

New line traffic = Robot + Sensors + CCTV + Office

= 32 + 0.24 + 4.8 + 30 = 67.04 Mbps

For eight production lines: 67.04 × 8 = 536.32 Mbps

Therefore, the total plant traffic becomes 536.32 Mbps.

Comparison

Parameter	Continuous CCTV	Local recording
CCTV traffic per line	96 Mbps	4.8 Mbps
Total line traffic	158.24 Mbps	67.04 Mbps
Eight-line traffic	1265.92 Mbps	536.32 Mbps
1 Gbps uplink	Insufficient	Sufficient

Local recording significantly reduces network traffic and makes the 1 Gbps uplink sufficient for the calculated load. However, local storage capacity and access to recorded footage must also be considered.

Conclusion- The traffic analysis shows that CCTV generates the highest data volume, whereas robot controllers require the lowest latency. A 1 Gbps uplink is sufficient for one production line but insufficient for all eight lines under continuous CCTV streaming. Implementing local CCTV recording, QoS and a higher-capacity backbone can improve network performance and reliability. The proposed design should prioritise industrial control traffic to ensure that robot commands reach their destinations within the required 10 milliseconds.

TASK 2 – COUNT THE DOMAINS IN THE CONTRACTOR'S DRAWING

1. Introduction

A network consists of different devices that communicate through physical and logical connections. Collision domains and broadcast domains help determine how efficiently a network operates. Understanding these domains is important when designing a network for a manufacturing plant.

The contractor proposes three hubs connected to a 24-port switch. The switch connects to a router through interface G0/0. Another switch, containing 12 IP phones, connects to the router through interface G0/1.

2. Calculation of Collision and Broadcast Domains

A. Collision Domains

A collision domain is a network segment in which devices share the same communication medium. When two devices transmit simultaneously on a shared half-duplex Ethernet medium, collisions may occur.

- Each hub connects 8 PCs. Since a hub repeats signals to all connected devices, all 8 PCs share one collision domain.
- Three hubs therefore create 3 collision domains.
- The connection between the first switch and router creates 1 separate collision domain.
- The second switch connects 12 IP phones. Each switch port creates an individual collision domain, resulting in 12 collision domains.
- The connection between the second switch and router creates 1 additional collision domain.

Total collision domains: 3 + 1 + 12 + 1 = **17 collision domains**

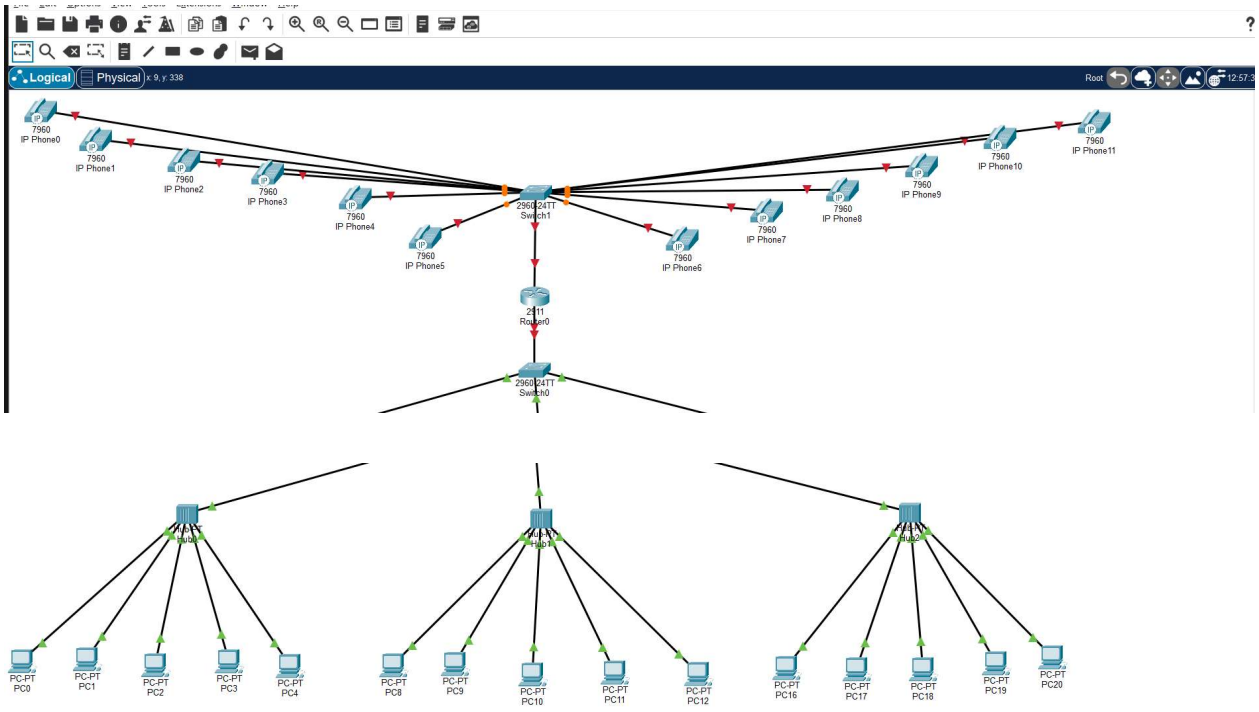
B. Broadcast Domains

A broadcast domain is a network segment in which broadcast messages can reach all devices without crossing a Layer 3 routing boundary. A router separates

broadcast domains because it does not normally forward Layer 2 broadcasts between interfaces. Since the router has two connected interfaces, G0/0 and G0/1:

Total broadcast domains = **2**

3. Network Diagram



The three hub segments belong to the first broadcast domain, while the second switch and its phones belong to the second broadcast domain.

4. OSI Layers of Network Devices

Device	OSI Layer	Function
Hub	Layer 1 – Physical	Repeats incoming electrical signals to connected ports.
Switch	Layer 2 – Data Link	Uses MAC addresses to forward Ethernet frames.
Router	Layer 3 – Network	Uses IP addresses to route packets between networks.

A hub cannot prevent collisions because it does not examine MAC addresses or maintain a forwarding table. It simply repeats incoming signals to all ports. A

switch separates collision domains by providing dedicated communication links to individual ports.

5. Point-to-Point and Multipoint Connections

Multipoint connections: The connections between PCs and hubs are multipoint because multiple devices share the same communication medium.

Point-to-point connections: Switch-to-router connections and individual switch-to-phone connections are point-to-point because each link connects two devices directly. Point-to-point links provide better communication isolation and reduce the possibility of collisions.

6. Network Standards and Standardisation Bodies

Network standards ensure that devices from different manufacturers can communicate correctly.

Technology	Standard	Standardisation body
Ethernet	IEEE 802.3	IEEE
Wi-Fi	IEEE 802.11	IEEE
IPv4	RFC 791	IETF
OSI Model	ISO 7498	ISO
5G	3GPP specifications / IMT-2020 framework	3GPP / ITU

IEEE develops standards for Ethernet and wireless LAN technologies. IETF develops Internet protocol standards through RFC documents. ISO defines international standards such as the OSI reference model. 3GPP develops mobile communication specifications, while ITU defines the IMT-2020 framework for 5G. The plant should purchase standard-compliant equipment because it provides interoperability, compatibility, reliability and easier maintenance. It also reduces

the risk of communication failures when equipment from different vendors is connected.

7. Alternative Concept Analysis – Replacing Hubs with Switches

A colleague suggests replacing the three hubs with three unmanaged switches.In the modified design:

- 24 individual PC connections create 24 collision domains.
- Three uplinks from the switches create 3 additional collision domains.
- The router connection creates 1 collision domain.
- The second switch continues to have 13 collision domains (12 phone links and 1 router link).

Total collision domains: = 24 + 3 + 1 + 12 + 1 = **41 collision domains**

Broadcast domains remain **2**, because unmanaged Layer 2 switches do not normally separate broadcast traffic.

Comparison

Parameter	Original design	Modified design
Collision domains	17	41
Broadcast domains	2	2
Main devices	Hubs and switches	Switches only
Collision management	Shared hub segments	Separate switch ports

Replacing hubs with switches reduces collisions because every switch port provides a separate collision domain. However, it does not eliminate broadcast traffic. VLANs and routing are required for further broadcast-domain separation.

Conclusion- The contractor's original network contains 17 collision domains and 2 broadcast domains. Hubs operate at the Physical Layer and cannot separate collisions, whereas switches provide dedicated collision domains. Replacing hubs with switches increases the number of collision domains to 41 and improves network performance. Using standard-compliant devices ensures compatibility and reliable communication throughout the manufacturing plant.

TASK 3 – READ A CAPTURED FRAME LAYER BY LAYER

1. Introduction

Data communication involves several layers that add control information before transmitting data across a network. This process is known as encapsulation. At the receiving device, the corresponding headers are removed through decapsulation. The given packet capture contains Ethernet II, IPv4, TCP and application data. Analysing these fields helps identify the source, destination, protocol and possible communication errors.

2. Identification of OSI Layers and PDUs

The captured frame contains the following information:

- Ethernet II: Source MAC and destination MAC addresses, EtherType 0x0800.
- IPv4: Source IP 10.10.1.25, destination IP 10.10.9.4, TTL 61.
- TCP: Source port 51522, destination port 502, sequence number 1201 and acknowledgement number 88.
- Data: 12 bytes containing a Modbus write-register command.

Captured field	OSI layer	PDU
Application data	Layer 7	Data
TCP	Layer 4	Segment
IPv4	Layer 3	Packet

Ethernet II	Layer 2	Frame

The EtherType 0x0800 indicates that the Ethernet frame carries an IPv4 packet. TCP protocol number 6 identifies the transport protocol. Port 502 is commonly used by Modbus TCP communication.

3. Calculation of Routers Crossed

The initial TTL value is 64, and the captured packet shows a TTL of 61.

Number of routers crossed = Initial TTL − Remaining TTL

= 64 − 61 = **3 routers**

Each router normally decreases the TTL value by one before forwarding the IPv4 packet. This prevents packets from circulating indefinitely.

4. Calculation of Total Bytes on the Wire

The packet contains 12 bytes of application data. The standard header sizes used for this calculation are:

- Ethernet header = 14 bytes
- IPv4 header = 20 bytes
- TCP header = 20 bytes
- Ethernet FCS trailer = 4 bytes

Total overhead = 14 + 20 + 20 + 4 = 58 bytes

Total frame size = 12 + 58 = **70 bytes**

Overhead percentage = (58 / 70) × 100 = **82.86%**

Therefore, most of the transmitted bytes are protocol overhead rather than actual application data.

Calculation for a 1,400-Byte Camera Packet

Application data = 1,400 bytes

Total overhead = 58 bytes

Total frame size = 1400 + 58 = **1458 bytes**

Overhead percentage = (58 / 1458) × 100 = **3.98%**

Comparison

Parameter	Robot command	Camera packet
Application data	12 bytes	1400 bytes
Overhead	58 bytes	58 bytes
Total size	70 bytes	1458 bytes
Overhead percentage	82.86%	3.98%

This comparison shows that small robot commands have proportionally higher protocol overhead. However, their small size makes them suitable for frequent communication when low latency is maintained.

5. Identifying Network Faults at Different Layers

A. Wrong MAC address – Layer 2

The Data Link Layer handles physical addressing using MAC addresses. An incorrect destination MAC address may prevent the Ethernet frame from reaching the intended device on the local network.

B. TTL reaches zero – Layer 3

The Network Layer handles IP addressing and routing. When TTL reaches zero, the router discards the packet and may generate an ICMP Time Exceeded message.

C. Destination port is closed – Layer 4

The Transport Layer manages TCP and UDP communication through port numbers. If a TCP destination port is closed, the destination system may respond with a TCP RST packet.

6. Mapping the OSI Model to TCP/IP Model

The TCP/IP model contains four layers and combines some functions of the seven-layer OSI model.

OSI Model	TCP/IP Model
Application	Application
Presentation	Application
Session	Application
Transport	Transport
Network	Internet
Data Link	Network Access
Physical	Network Access

The OSI model is mainly used as a conceptual reference for understanding network communication. The TCP/IP model represents the practical architecture used by the Internet.

7. Complete the Code – Encapsulation and Decapsulation

The following Python program demonstrates how each layer adds a header and how the receiver removes the headers.

```
File  Edit  Format  Run  Options  Window  Help
"""How data is wrapped and unwrapped."""
LAYERS = [("Application", "HTTP"), ("Transport", "TCP"), ("Network", "IP"), ("Data link", "ETH")]

def send(message):
    pdu = message
    for layer, tag in LAYERS:
        tail = "|FCS]" if tag == "ETH" else "]"
        pdu = "[" + tag + "|" + pdu + tail
        print(f"{layer} adds {tag} -> {len(pdu)} bytes")
    return pdu

def receive(frame):
    pdu = frame
    for layer, tag in reversed(LAYERS):
        head = "[" + tag + "|"
        tail = "|FCS]" if tag == "ETH" else "]"
        assert pdu.startswith(head) and pdu.endswith(tail)
        pdu = pdu[len(head):-len(tail)]
    return pdu

frame = send("GET /results")
print("on the wire:", frame)
print("delivered :", receive(frame))
```

Expected Output:

```
===== RESTART: C:/Users/anand/OneDrive/Documents/Python/CN Assignment-1.py =
====
Application adds HTTP -> 19 bytes
Transport adds TCP -> 25 bytes
Network adds IP -> 30 bytes
Data link adds ETH -> 40 bytes
on the wire: [ETH|[IP|[TCP|[HTTP|GET /results]]]|FCS]
delivered : GET /results
overhead : 28 bytes for 12 bytes of data
>>>
```

Changing the Message to 1,400 Bytes

The simplified program adds 28 bytes of overhead.

For 12 bytes: Overhead relative to data = (28/12) × 100 = 233.33%.

For 1,400 bytes:Overhead relative to data = (28/1400) × 100 = 2%.

Thus, the percentage of overhead decreases significantly as the application data size increases.

8. Alternative Concept Analysis – Direct Ethernet Communication

An industrial protocol could transmit robot commands directly inside an Ethernet frame using EtherType 0x8892 without IPv4 and TCP headers.

IP header = 20 bytes

TCP header = 20 bytes

Total bytes saved per command= 20 + 20 = **40 bytes**

On a 100 Mbps link:

Number of additional commands supported per second = 100,000,000 / (40 × 8)

= **312,500 commands per second**

Direct Ethernet communication reduces protocol overhead and can improve efficiency for industrial control. However, removing IP means losing IP-based addressing and routing. Removing TCP also means losing TCP's built-in acknowledgement and retransmission mechanisms. Therefore, the industrial protocol must provide suitable communication and reliability mechanisms where required.

Conclusion- Encapsulation allows network layers to add the information required for successful communication. The captured packet crossed three routers, and its small application payload resulted in a high overhead percentage. The Python program demonstrates encapsulation and decapsulation, while direct Ethernet communication can reduce overhead for industrial commands. The final choice should balance efficiency, routing requirements and reliability.

TASK 4 – CHOOSE A MEDIUM FOR EACH RUN

1. Introduction

Transmission media provide the physical or wireless path through which data travels between network devices. Selecting the correct medium is essential in a smart manufacturing plant because different locations have different distance, bandwidth and interference requirements. Electrical noise from welding machines

can affect copper communication, while long-distance connections require media that support reliable transmission over greater distances.

2. Selection of Transmission Medium

The plant contains four cable runs with different requirements.

Cable run	Distance	Recommended medium	Reason
Robot cell near welders	85 m	Cat6A S/FTP	Shielding helps reduce electromagnetic interference.
Office to shop-floor switch	140 m	Multimode fibre	Exceeds the normal 100 m copper Ethernet channel limit.
Shop floor to data centre	550 m	Single-mode fibre	Supports reliable longer-distance communication.
Plant to head office	400 km	Single-mode fibre leased line	Suitable for long-distance, high-capacity connectivity.

Robot Cell – 85 m: Cat6A S/FTP is selected because it provides shielding against external electrical interference. Proper grounding and installation are also important.

Office to Shop Floor – 140 m: Copper Ethernet normally supports a maximum channel length of approximately 100 m. Therefore, fibre is selected to avoid exceeding this limit.

Shop Floor to Data Centre – 550 m: Single-mode fibre is suitable for this distance and supports future bandwidth expansion.

Plant to Head Office – 400 km: A single-mode fibre leased service is appropriate because it supports long-distance communication and high data rates.

3. Identification of the Copper Distance Limitation

Traditional twisted-pair Ethernet connections generally have a maximum channel length of 100 metres.The office-to-shop-floor connection is 140 metres, the data-centre connection is 550 metres, and the head-office connection is 400 kilometres.Therefore, these three runs cannot use a single ordinary copper Ethernet channel within its standard distance limit.Fibre provides a better solution because it supports longer distances and is not affected by electromagnetic interference in the same way as copper.

4. Shannon Capacity Calculation

Shannon's theorem determines the theoretical maximum data rate of a communication channel based on its bandwidth and signal-to-noise ratio.

Formula:

[C=B\log_2(1+SNR)]

Where:

- C = Channel capacity in bits per second
- B = Bandwidth in hertz
- SNR = Signal-to-noise ratio as a linear value

Given: Bandwidth = 250 MHz

[B=250\times10^6\ Hz]

A. Shannon Capacity at 40 dB SNR

First, convert decibels into a linear ratio:

[SNR=10^{40/10}]

[SNR=10,000]

Substituting into Shannon's formula:

[C=250\times10^6\log_2(1+10000)]

[C=250\times10^6\times13.2879]

[\boxed{C=3321.96\ Mbps}]

Therefore, the theoretical Shannon capacity at 40 dB SNR is approximately 3.322 Gbps.

B. Shannon Capacity at 15 dB SNR

Convert the SNR:

[SNR=10^{15/10}]

[SNR=31.62]

Substituting:

[C=250\times10^6\log_2(1+31.62)]

[C=250\times10^6\times5.028]

[\boxed{C\approx1256.95\ Mbps}]

Comparison of Both Results

Parameter	40 dB SNR	15 dB SNR
Bandwidth	250 MHz	250 MHz
SNR	10,000	31.62
Shannon capacity	3321.96 Mbps	1256.95 Mbps

The reduction in SNR decreases the theoretical channel capacity considerably. Near welding machines, electromagnetic interference can reduce signal quality and increase transmission errors. Therefore, shielded cabling or fibre is preferred for critical industrial communication.

5. Why Actual Link Speed Is Below Shannon Capacity

Shannon capacity represents a theoretical upper limit under ideal coding assumptions. Real networks experience several limitations:

1. **Noise and interference:** External signals can affect data transmission.
2. **Hardware limitations:** Network cards, switches and transceivers have maximum operating speeds.
3. **Protocol overhead:** Ethernet, IP and TCP headers consume part of the available bandwidth.
4. **Signal attenuation:** Signal strength decreases over distance.
5. **Transmission errors:** Retransmissions and error correction reduce useful throughput.

Therefore, the actual data rate is normally lower than the theoretical Shannon capacity.

6. Complete the Code – Nyquist and Shannon Limits

The following Python program calculates the Nyquist maximum bit rate and Shannon channel capacity.

```
"""Nyquist and Shannon limits."""
import math

def nyquist(bandwidth_hz, levels):
    return 2 * bandwidth_hz * math.log2(levels)

def shannon(bandwidth_hz, snr_db):
    snr = 10 ** (snr_db / 10)
    return bandwidth_hz * math.log2(1 + snr)

media = [
    ("Telephone line", 3100, 30),
    ("Cat6 pair", 250e6, 40),
    ("Wi-Fi 6 channel", 160e6, 25)
]

for name, bw, snr in media:
    print(f"{name:<16} Shannon limit "
          f"{shannon(bw, snr) / 1e6:10.2f} Mbps")

print(f"Nyquist, 3,100 Hz with 4 levels: "
      f"{nyquist(3100, 4) / 1e3:.1f} kbps")
```

Expected Output:

```
===== RESTART: C:/Users/anand/OneDrive/Documents/Python/CN Assignment-1.py =
====
Telephone line   Shannon limit      0.03 Mbps
Cat6 pair        Shannon limit   3321.96 Mbps
Wi-Fi 6 channel  Shannon limit   1329.50 Mbps
Nyquist, 3,100 Hz with 4 levels: 12.4 kbps
>>>
```

Telephone Line SNR Required for 56 kbps

Given:Bandwidth = 3100 Hz and Required capacity = 56,000 bps

Using Shannon's formula:

[56000=3100\log_2(1+SNR)]

[\frac{56000}{3100}=\log_2(1+SNR)]

[SNR=2^{56000/3100}-1]

[SNR\approx274,000]

Converting into decibels:

[SNR_{dB}=10\log_{10}(274000)]

[\boxed{SNR\approx54.38\ dB}]

Such a high SNR requirement is unrealistic for an ordinary telephone line, making 56 kbps transmission difficult under these assumptions.

7. Alternative Concept Analysis – Wi-Fi 6 Near Welding Machines

Suppose the robot cell uses Wi-Fi 6 instead of shielded cable.

Given: Bandwidth = 160 MHz ;SNR = 15 dB

[SNR=10^{15/10}=31.62]

Using Shannon's formula:

[C=160\times10^6\log_2(1+31.62)]

[C\approx160\times10^6\times5.028]

[\boxed{C\approx804.45\ Mbps}]

The theoretical capacity is approximately 804.45 Mbps. However, high theoretical bandwidth does not guarantee that robot commands will arrive within 10 milliseconds. Wireless interference, shared-channel contention, retransmissions and variable latency can affect real-time communication. Therefore, Wi-Fi 6 should not be the sole communication medium for critical robot control. Shielded industrial Ethernet or an appropriately engineered private 5G URLLC solution is more suitable.

Conclusion- The transmission medium must be selected according to distance, interference and performance requirements. Shielded Cat6A is suitable for shorter industrial cable runs, while fibre is preferred for longer connections. Shannon's theorem shows that channel capacity decreases when SNR falls. Although Wi-Fi 6 provides high theoretical capacity, it cannot independently guarantee the strict latency requirements of robot control. A combination of shielded cabling and fibre provides a more reliable foundation for the manufacturing plant network.

TASK 5 – Wireless Technologies and Mobility

1 AGV Mobility and Handover Calculation

An Automated Guided Vehicle (AGV) travels along a 1.2 km route at 2 m/s.

Given:

- Route length = 1200 m
- Speed = 2 m/s
- Wi-Fi access point spacing = 30 m
- Wi-Fi handover delay = 150 ms
- Private 5G cell coverage = 300 m
- 5G handover delay = 20 ms

Travel time: Travel time = Distance / Speed = 1200 / 2 = 600 seconds (10 minutes)

Wi-Fi handovers:

Number of coverage sections = 1200 / 30 = 40

Assuming 40 coverage sections, handovers = 39.

Total interruption = 39 × 150 ms = 5850 ms = **5.85 seconds**.

5G handovers:

Number of cells = 1200 / 300 = 4

Handovers = 4 − 1 = 3.

Total interruption = 3 × 20 ms = **60 ms**.

Therefore, private 5G provides significantly lower handover interruption than the given Wi-Fi arrangement. However, even a 20 ms handover exceeds the AGV's strict 10 ms command deadline. Local control and seamless connectivity mechanisms are still necessary.

2 Wireless Technology Selection

Application	Suitable technology	Reason
AGVs	URLLC / Private 5G	Low latency and reliable communication
CCTV cameras	eMBB	Requires high data throughput
Large sensor networks	mMTC	Supports many connected devices
Maintenance peripherals	Bluetooth	Short-range communication
Sensor mesh	Zigbee	Low-power communication
Remote low-rate sensors	LoRa	Long-range communication

3 Wi-Fi 6 Fast Roaming

IEEE 802.11r supports fast BSS transition, reducing authentication delays during Wi-Fi roaming. It can improve AGV mobility, but does not guarantee that the complete handover will meet a 10 ms deadline. Possible risks include interference, packet loss, network congestion and authentication issues.

4 LoRa Sensor Capacity

Given:

- Number of sensors = 600
- Data per sensor = 50 bytes/s
- LoRa gateway capacity = 5 kbps

Total sensor traffic: 600 × 50 × 8 = 240,000 bps = **240 kbps**.

Traffic per sensor: 50 × 8 = 400 bps.

Number of sensors supported: 5000 / 400 = 12.5.

Therefore, approximately **12 sensors** can be supported theoretically by one 5 kbps gateway. Actual capacity may be lower due to protocol overhead and transmission restrictions. The remaining sensors should use suitable wired, Wi-Fi or other industrial communication networks depending on their data rate and latency requirements.

Conclusion: Private 5G is more suitable for mobile AGVs requiring reliable low-latency communication, while Wi-Fi, Zigbee and LoRa can support other applications according to their requirements.

TASK 6 – Network Troubleshooting and Ethernet Technologies

1 Identifying the Network Problem

The shop-floor PC reports the following:

Parameter	Observed value
Speed	100 Mbps
Duplex	Half
Auto-negotiation	Off
Link	Yes
RX errors	2381

Late collisions	412
CRC errors	1907

The high number of late collisions and CRC errors indicates a communication problem. A possible cause is a duplex mismatch between the PC and switch. A late collision occurs when a collision is detected after the normal Ethernet collision -detection window. CRC errors indicate that received frames have failed their error-checking test.

2 Corrective Action

The recommended solution is to enable auto-negotiation on both the PC and switch. On Linux, the following command can be used:

```
None
sudo ethtool -s eth0 autoneg on
```

Verify the configuration using:

```
None
ethtool eth0
```

On a Cisco switch, configure the relevant interface:

```
None
interface fa0/1
speed auto
duplex auto
```

Both ends should use compatible settings. After correction, check whether error counters stop increasing.

3 MAC Address Identification

Given MAC address: 01:00:5e:0a:01:02

- The prefix 01:00:5e identifies an IPv4 multicast Ethernet MAC address range.
- The address is used for multicast communication rather than identifying a single ordinary host.
- The first three bytes 3c:52:82 of another address can be used as an OUI to identify its registered manufacturer.

4 Jumbo Frames

Jumbo frames carry larger Ethernet payloads, commonly around 9000 bytes instead of the standard 1500-byte MTU.

Advantages:

- Reduces the number of packets transmitted.
- Reduces per-packet processing overhead.
- Can improve efficiency when transferring large CCTV recordings.

All devices along the communication path must support the configured jumbo MTU. Otherwise, packets may be dropped or require fragmentation at higher network layers.

5 Alternative Network Topology: Ring

An industrial ring topology provides an alternative communication path if one link fails. With a suitable ring recovery protocol, communication can resume through the alternate path. A star topology is simpler to manage, but failure of its central switch can disconnect all devices attached to it. Ring recovery time depends on the protocol and equipment used.

Conclusion: Correct duplex configuration, error monitoring and suitable Ethernet frame sizes improve network reliability. Ring redundancy can provide additional fault tolerance in industrial environments.

TASK 7 – Error Detection and Correction

1 CRC Calculation

CRC (Cyclic Redundancy Check) detects transmission errors using binary polynomial division.

Given:

- Data = 1101011011
- Generator = 1011
- Generator degree = 3

Append three zeros to the data: 1101011011000

Perform modulo-2 division using XOR operations.

CRC remainder = **100**

Therefore, the transmitted frame is: 1101011011100

If the received frame is 1101011011101, its remainder is non-zero. Hence, the receiver identifies an error and rejects the frame.

2 Burst Error Detection

The generator has degree 3. Therefore, CRC guarantees detection of all burst errors of length 3 bits or less. Longer burst errors may also be detected, but detection is not guaranteed.An error pattern that is a multiple of the generator polynomial can remain undetected. For example:

1011000 = 1011 × 1000

This error pattern is divisible by the generator and can therefore escape CRC detection.

3 Two-Dimensional Parity

Consider the following 4 × 4 data block:

```
None
1 0 1 0 | 0
0 1 1 0 | 0
1 1 0 0 | 0
0 0 1 1 | 0
----------
0 0 1 1 | 0
```

Even parity is used for each row and column. The final row contains column parity bits, and the last bit is the overall parity. Two-dimensional parity can identify many errors by locating the row and column containing an incorrect parity value. However, if four bits at the corners of a rectangle are changed, each affected row and column still has an even number of errors. This pattern can escape detection.

4 Internet Checksum

Given 16-bit words:

- 4500
- 0073
- 0000
- 4000

Add the values: 4500 + 0073 + 0000 + 4000 = 8573

Take the one's complement: FFFF − 8573 = 7A8C

Therefore, the Internet checksum is **7A8C**.

5 Python Program for CRC

The following program performs modulo-2 division and checks the received data.

```
def mod2_div(dividend, divisor):
    rem = list(dividend)
    for i in range(len(dividend) - len(divisor) + 1):
        if rem[i] == "1":
            for j in range(len(divisor)):
                rem[i+j] = str(
```

```
            rem[i+j] = str(
                int(rem[i+j]) ^ int(divisor[j])
            )
    return "".join(rem[-(len(divisor)-1):])

def encode(data, divisor):
    padded = data + "0" * (len(divisor)-1)
    crc = mod2_div(padded, divisor)
    return data + crc

def check(frame, divisor):
    return mod2_div(frame, divisor)

data = "1101011011"
divisor = "1011"

frame = encode(data, divisor)
print("CRC:", frame[-3:])
print("Transmitted frame:", frame)

received = "1101011011101"
remainder = check(received, divisor)
```

Expected output:

```
===== RESTART: C:/Users/anand/OneDrive/Documents/Python/CN Assignment-1.py =
====
CRC: 100
Transmitted frame: 1101011011100
Received remainder: 001
Error detected
```

6 Comparison of Even Parity and CRC-3

Error pattern	Even parity	CRC-3
0000011 (2-bit burst)	Not detected	Detected
00001111 (4-bit burst)	Not detected	Detected
1011000	Detected	Not detected

Even parity detects an odd number of bit changes but misses errors involving an even number of changed bits. CRC-3 detects all burst errors of length up to three bits, but some longer patterns divisible by its generator can escape detection.

Conclusion: CRC provides stronger error detection than simple parity for many communication errors. Two-dimensional parity and Internet checksum provide additional methods for detecting corrupted data.

TASK 8 – Correct the Error or Ask Again?

1 Hamming Check Bits

A vibration sensor sends 16-bit readings over a noisy radio link.

Given: m = 16 data bits.

Using the formula: $2^r \geq m + r + 1$

For r = 5: $32 \geq 16 + 5 + 1 = 22$

Therefore, **5 check bits** are required.

Total transmitted bits = 16 + 5 = 21 bits.

Overhead percentage = (5 / 16) × 100 = **31.25%**.

2 Minimum Hamming Distance

Given codewords: 00000, 01011, 10101, 11110

The minimum Hamming distance between any two codewords is **3**.

- Error detection = d − 1 = 2 bits.
- Error correction = (d − 1) / 2 = 1 bit.

Therefore, the code can detect up to 2 errors and correct 1 error.

3 Correcting the Received Codeword

Original codeword = 0110011

Received codeword = 0110111

Comparing both, bit position 5 is incorrect.

After correcting bit 5:

Corrected codeword = 0110011

Extracting data bits from positions 3, 5, 6 and 7 gives:

Corrected data = 1011

4 Forward Error Correction vs ARQ

Forward Error Correction (FEC) corrects errors using additional check bits without requesting retransmission. ARQ detects errors and requests another transmission.

Given:

- Data size = 16 bits
- Bit error rate = 1/1000
- Retransmission time = 40 ms

Probability of an erroneous reading:

$P = 1 - (0.999)^{16}$

$P \approx 0.01588 = 1.588\%$

FEC:

- Bits transmitted = 21 bits.
- Corrects a single-bit error without retransmission.
- Avoids the additional 40 ms retransmission wait.

ARQ:

- Average transmitted bits ≈ 16 / (1 − 0.01588) = 16.26 bits.

- Average additional retransmission delay ≈ 0.01588 × 40 = 0.635 ms.

FEC is preferable when predictable delivery time is important. Video streaming and 5G use FEC because retransmissions can increase delay and interrupt real-time communication.

5 Python Program – Hamming(7,4)

```
def encode(d):
    p1 = d[0] ^ d[1] ^ d[3]
    p2 = d[0] ^ d[2] ^ d[3]
    p3 = d[1] ^ d[2] ^ d[3]
    return [p1, p2, d[0], p3, d[1], d[2], d[3]]

def decode(c):
    s1 = c[0] ^ c[2] ^ c[4] ^ c[6]
    s2 = c[1] ^ c[2] ^ c[5] ^ c[6]
    s3 = c[3] ^ c[4] ^ c[5] ^ c[6]
    pos = s1 + 2*s2 + 4*s3
    if pos != 0:
        c[pos-1] ^= 1
    data = [c[2], c[4], c[5], c[6]]
    return pos, data
word = encode([1, 0, 1, 1])
print("codeword:", "".join(map(str, word)))
for flip in (None, 5):
    rx = word[:]
    if flip:
        rx[flip-1] ^= 1
    pos, data = decode(rx)
    print("error at", flip, "syndrome", pos,
```

Expected output:

```
===== RESTART: C:/Users/anand/OneDrive/Documents/Python/CN Assignment-1.py =
====
codeword: 0110011
error at None syndrome 0 data 1011
error at 5 syndrome 5 data 1011
>>>
```

If two bits (positions 5 and 6) are flipped, the decoder produces syndrome 3 and incorrectly changes bit 3. The resulting data becomes 0101 instead of 1011. This demonstrates that ordinary Hamming(7,4) cannot reliably correct multiple-bit errors.

Conclusion: Hamming coding provides efficient single-bit error correction. FEC is suitable for noisy links where retransmission delays are undesirable.

TASK 9 – How Big Must the Window Be?

1 Transmission and Propagation Time

Given:

- Distance = 400 km = 400,000 m
- Link speed = 1 Gbps
- Frame size = 1500 bytes = 12,000 bits
- Propagation speed = 2×10^8 m/s

Transmission time: Tt = 12,000 / 1,000,000,000 = **0.000012 seconds = 12 µs**

Propagation time: Tp = 400,000 / (2×10^8) = **0.002 seconds = 2 ms**

Ratio:a = Tp / Tt = 0.002 / 0.000012 = **166.67**

2 Stop-and-Wait Utilisation

Utilisation = 1 / (1 + 2a) = 1 / (1 + 2 × 166.67) ≈ **0.299%**

Throughput = 0.00299 × 1000 Mbps ≈ **2.99 Mbps**

Therefore, Stop-and-Wait uses the 1 Gbps link very inefficiently.

3 Required Sliding Window

For full utilisation: $W \geq 1 + 2a$

$W \geq 334.33$

Therefore, the smallest integer window is **335 frames**.

Sequence number bits:

- Go-Back-N: $2^n - 1 \geq 335$, so **9 bits**.
- Selective Repeat: $2^{n-1} \geq 335$, so **10 bits**.

4 Go-Back-N vs Selective Repeat

Feature	Go-Back-N	Selective Repeat
Retransmission	Lost frame and following frames	Only lost frames
Receiver	Simpler	Requires buffering
Efficiency	Lower after packet loss	Higher after packet loss

Selective Repeat requires more receiver memory but avoids unnecessary retransmissions. For a 335-frame window:

Buffer space = 335 × 1500 = **502,500 bytes**.

5 HDLC and PPP Framing

HDLC and PPP use a flag pattern 01111110 to mark the beginning and end of a frame. Bit stuffing prevents the flag pattern from appearing accidentally within the data. A zero is inserted after five consecutive 1s and removed by the receiver.

6 Alternative: Selective Repeat with Window 64

Given window W = 64: Utilisation = 64 / (1 + 2 × 166.67) ≈ **19.14%**

Throughput = 0.1914 × 1000 ≈ **191.4 Mbps**

Receiver buffer: 64 × 1500 = **96,000 bytes**.

This is better than Stop-and-Wait but does not fully utilise the 1 Gbps link.

7 Python Program – Link Utilisation

```
def utilisation(frame_bits, rate_bps, distance_m,
            window, speed=2e8):
    tx = frame_bits / rate_bps
    prop = distance_m / speed
    a = prop / tx
    u = min(1, window / (1 + 2*a))
    return u, a
links = [("Campus LAN, 200 m, 1 Gbps", 1e9, 200), ("City fibre, 50 km, 1 Gbps", 1e9, 50000),("Satellite, 36,000 km, 10 Mbps", 10e6, 3

for name, rate, dist in links:
    u1, a = utilisation(12000, rate, dist, 1)
    u7, _ = utilisation(12000, rate, dist, 7)
    u127, _ = utilisation(12000, rate, dist, 127)
    print(f"{name:<31} a = {a:7.2f} "
          f"stop-and-wait {u1:6.1%} "
          f"W=7 {u7:6.1%} W=127 {u127:6.1%}")
```

Expected output:

```
===== RESTART: C:/Users/anand/OneDrive/Documents/Python/CN Assignment-1.py =
====
Campus LAN, 200 m, 1 Gbps       a =    0.08 stop-and-wait  85.7% W=7 100.0% W=127 100.0%
City fibre, 50 km, 1 Gbps       a =   20.83 stop-and-wait   2.3% W=7  16.4% W=127 100.0%
Satellite, 36,000 km, 10 Mbps   a =  150.00 stop-and-wait   0.3% W=7   2.3% W=127  42.2%
>>>
```

Conclusion: Sliding-window protocols provide better utilisation on long-distance links. Selective Repeat is more efficient after packet loss, while Go-Back-N requires simpler receiver buffering.

TASK 10 – Find Where the Delay Comes From

1 Fibre Propagation Delay

Given:

- Distance = 400 km = 400,000 m
- Propagation speed = 2×10^8 m/s

One-way propagation delay: = 400,000 / (2×10^8) = **2 ms**

Round-trip propagation delay = 2 × 2 = **4 ms**

The final measured traceroute RTT is approximately 12.6 ms. Therefore, the theoretical fibre propagation accounts for about:

(4 / 12.6) × 100 = **31.75%**

The remaining delay comes from equipment processing, switching, queuing and other network paths.

2 Traceroute Analysis

Hop	Address	Average observation
1	10.10.1.1	0.6 ms
2	100.64.0.1	3.1 ms
3	172.31.4.9	4.0 ms
4	103.21.8.17	11.8 ms
5	No response	Timeout
6	10.200.0.10	12.6 ms

Hop 3 – 38.2 ms: One probe has a much higher response time than the other two. This may be caused by temporary congestion or delayed ICMP processing. It does not necessarily indicate a forwarding fault.

Hop 5 – Stars: The router may block or limit traceroute responses. Since hop 6 responds, hop 5's missing response does not prove that the network is disconnected.

Largest delay increase: Hop 4 adds approximately 7.8 ms compared with the normal hop 3 response of 4.0 ms. It is the largest apparent increase.

3 Monitoring Method

Method	Purpose
Ping	Checks reachability and round-trip time
Traceroute	Identifies the path and response delay at each hop
SNMP	Monitors interface traffic, errors and device status
NetFlow	Records traffic flows and helps identify heavy network users
Wireshark	Captures packets and examines protocol headers

Continuous SNMP and NetFlow monitoring can identify recurring congestion and traffic problems. Occasional ping and traceroute tests provide useful snapshots but may miss short-lived problems.

4 Practical Testing on Windows

Open Command Prompt and execute:

Traceroute:

```
None
tracert www.google.com
```

Ping:

```
None
ping www.google.com -n 10
```

For packet analysis, open Wireshark, select your active network interface and start capturing while performing the ping or traceroute. The captured packets can be examined at these layers:

- Data Link: Ethernet header and MAC addresses.
- Network: IPv4 header and source/destination IP addresses.

- Transport: ICMP for Windows ping/traceroute probes, or TCP/UDP for other traffic.

Take screenshots of your actual command outputs and Wireshark capture as evidence.

Conclusion: Traceroute helps identify where network delay increases, while ping measures reachability and round-trip time. SNMP, NetFlow and Wireshark provide deeper monitoring and help locate recurring network problems.

TASK 11 – Defend the Network Design

1 Recommended Network Design

A hybrid network combining wired Ethernet, fibre, Wi-Fi 6 and private 5G is recommended for the smart manufacturing plant.

Area / Application	Recommended medium or technology	Reason
Robot cells	Cat6A S/FTP and private 5G URLLC	Shielding and low-latency communication
AGVs	Private 5G URLLC	Mobility and reliable handovers
CCTV cameras	Wired Ethernet and fibre backbone	Continuous high-volume video
Sensors	mMTC / Zigbee / LoRa where suitable	Supports large numbers of low-power devices
Office users	Wi-Fi 6	High-speed wireless access
Plant to head office	Single-mode fibre leased connection	Long-distance communication

Ethernet connections should preferably use full-duplex mode with auto-negotiation enabled at both ends. VLANs and Quality of Service (QoS) should separate robot, CCTV, sensor and office traffic.

2 Error Control and Flow Control

Link / Application	Recommended method	Purpose
Ethernet	CRC-32	Detects corrupted frames
TCP applications	ACK and retransmission	Reliable delivery
Wireless links	FEC and HARQ	Detects and corrects transmission errors
Robot commands	Sequence numbers, timeouts and safe-state control	Prevents outdated or missed commands from causing unsafe operation
Network traffic	QoS and priority queues	Gives time-critical traffic higher priority

From Task 1, the plant requires approximately 1.266 Gbps for eight production lines. Therefore, a single 1 Gbps uplink is insufficient. With motion-triggered CCTV recording, the estimated traffic reduces to approximately 536.32 Mbps, making a 1 Gbps link sufficient under the stated traffic assumptions.

3 Major Risks

1. **Network congestion:** CCTV and office traffic may delay time-critical robot commands if QoS and traffic separation are not configured properly.
2. **Wireless interference and handover:** Welding equipment can affect wireless communication, while moving AGVs may experience temporary connectivity interruptions.

These risks can be reduced through shielded cables, fibre, private 5G, redundant paths and continuous monitoring.

4 Expansion to Five Plants

One plant generates approximately 1.266 Gbps.

For five plants: 5 × 1.26592 = **6.3296 Gbps**

Therefore, the head-office backbone should provide multi-gigabit capacity, preferably a suitably designed 10 Gbps or higher backbone, with sufficient WAN capacity for actual peak traffic.

5 Alternative: Entire Plant Using Private 5G

If the entire plant uses private 5G, the technology can carry mobile AGV and robot communication through URLLC and suitable sensor traffic through mMTC. However, continuous CCTV traffic requires substantial bandwidth, and wired Ethernet and fibre provide predictable capacity without wireless interference. Therefore, CCTV storage and the main backbone should remain wired.

Conclusion: A hybrid network is the most suitable solution because it combines the reliability of wired Ethernet and fibre with the flexibility of private 5G and Wi-Fi 6.

TASK 12 – Re-evaluate and Adapt the Design

1 A. Impact Analysis

The company introduces a private 5G network and 50 AGVs that require reliable command delivery. The original network must be modified to support additional mobile devices and low-latency communication.

Assumption for calculation: Since the assignment does not specify AGV command size or frequency, each AGV is assumed to generate the same traffic as one robot controller in Task 1: a 200-byte command every 2 ms.

Traffic per AGV: 200 × 8 / 0.002 = 800,000 bps = 0.8 Mbps.

Traffic for 50 AGVs: 50 × 0.8 = **40 Mbps**.

Original plant traffic: 8 × 158.24 = 1265.92 Mbps.

Updated plant traffic: 1265.92 + 40 = **1305.92 Mbps (approximately 1.306 Gbps)**.

Thus, the additional AGVs increase the plant's traffic by 40 Mbps under the stated assumption.

Changes required:

- Introduce private 5G for AGVs.
- Use URLLC for mobile equipment and robot control.
- Retain shielded Ethernet and fibre for fixed devices and backbone connections.
- Use suitable FEC and wireless retransmission mechanisms.
- Apply QoS to prioritise time-critical commands.

2 B. Test the Original Design

The original Wi-Fi arrangement has a 150 ms handover delay, while private 5G has a 20 ms handover delay. For one AGV travelling along the 1.2 km route:

Parameter	Wi-Fi	Private 5G
Coverage distance	30 m	300 m
Handover count	39	3
Interruption per handover	150 ms	20 ms
Total interruption	5850 ms	60 ms

The AGV command deadline is 10 ms.

Both handover delays exceed this deadline, but private 5G significantly reduces the interruption. For 50 AGVs, the network must also support simultaneous communication and handovers. The original ordinary Wi-Fi design cannot reliably guarantee the required command deadline.

3 C. Adaptation or Redesign

The recommended decision is **MODIFY** the original design rather than completely replace it. The revised network will use:

- Private 5G URLLC for AGVs and mobile robot communication.
- Shielded Cat6A Ethernet for fixed robot equipment.
- Wi-Fi 6 for office users.
- Wired Ethernet and fibre for CCTV and data-centre connections.
- Single-mode fibre for long-distance plant-to-head-office communication.
- QoS, traffic separation and redundant network paths.

Local control and safe-state mechanisms should also be provided so that AGVs do not depend entirely on uninterrupted wireless communication.

4 D. Evidence-Based Defence

The modified design is preferred for the following reasons:

1. Original plant traffic is approximately 1.266 Gbps, increasing to approximately 1.306 Gbps with 50 assumed AGV traffic streams.
2. Wi-Fi produces approximately 5.85 seconds of total interruption per AGV trip, whereas private 5G produces approximately 60 ms.
3. Private 5G reduces the number of handovers from 39 to 3.
4. Fibre and shielded Ethernet provide reliable connections for fixed equipment and high-volume traffic.
5. Wi-Fi 6 remains suitable for office users, avoiding unnecessary private 5G usage.

Although private 5G requires additional infrastructure and management, it improves mobility and reduces handover interruptions. Its 20 ms handover figure

still does not guarantee the 10 ms command deadline, so local control and additional reliability mechanisms remain essential.

Final Conclusion: The original network should be modified into a hybrid industrial network. Private 5G URLLC supports AGV mobility, while shielded Ethernet, fibre and Wi-Fi 6 serve fixed equipment, CCTV and office users. This design provides better scalability, reliability and communication performance for the smart manufacturing plant.